CascadeRoute-VL: Taming the Reasoning Tax in Vision-Language Models via Online Contextual Bandit Routing

Supplementary Material

This supplementary document provides: (i) complete proofs of Lemmas 1–2, Theorem 1 (with OOD Warm-Start prior decay), and Theorem 2 (Sec. S1); (ii) full 17-VLM profiles, unsupervised dataset perceptual diagnostics, compact model rescue breakdown, and complete 14×17 real outcome/token matrices across all 30,540 queries (Sec. S2); (iii) full Pareto sweep tables, paired statistical tests with Bonferroni correction, and 17-arm domain-specialized routing frequencies (Sec. S3); and (iv) expanded visual qualitative case studies and full reproducibility details (Sec. S4).

S1. Complete Theoretical Proofs

S1.1. Proof of Lemma 1 (Bounded Context Space)

Let the d=904 feature representation for query (I_t, Q_t) be

$$s_t = [\, w_q \hat{q}_t \parallel w_v \hat{v}_t \parallel w_p p_t \,] \in \mathbb{R}^{904},$$

where $\hat{q}_t \in \mathbb{R}^{384}$ is L_2-normalized ($\|\hat{q}_t\|_2 = 1$), $\hat{v}_t \in \mathbb{R}^{512}$ is L_2-normalized ($\|\hat{v}_t\|_2 = 1$), and $p_t \in [0,1]^8$ comprises eight bounded unsupervised cues: normalized luminance entropy $H(I_t)/8 \in [0,1]$, Sobel edge gradient fraction in $[0,1]$, logarithmic aspect ratio $\frac{1}{\pi}\arctan(\ln(W/H))+0.5 \in (0,1)$, clipped log pixel area $\text{clip}(\log_{10}(WH)/7, 0, 1) \in [0,1]$, bright background fraction $\mathbb{E}[L > 235/255] \in [0,1]$, clipped prompt word ratio $\min(w, 128)/128 \in [0,1]$, symbolic/MCQ cue density in $[0,1]$, and sentence count ratio $\min(s, 16)/16 \in [0,1]$.

Because the three concatenated blocks occupy disjoint coordinate subsets and each entry of p_t lies in $[0,1]$, $\|p_t\|_2^2 \le \sum_{j=1}^{8} 1^2 = 8$. Substituting weights $(w_q, w_v, w_p) = (0.50, 0.55, 0.235)$ yields:

$$\begin{aligned} \|s_t\|_2^2 &= w_q^2\|\hat{q}_t\|_2^2 + w_v^2\|\hat{v}_t\|_2^2 + w_p^2\|p_t\|_2^2 \\ &\le 0.50^2(1) + 0.55^2(1) + 0.235^2(8) \\ &= 0.2500 + 0.3025 + 0.4418 = 0.9943 < 1.0, \end{aligned} \quad (1)$$

so $\|s_t\|_2 \le \sqrt{0.9943} \approx 0.99715 < 1$. Across all 30,540 encoded real benchmark queries, we empirically verify $\min_t \|s_t\|_2 = 0.7611$, $\text{mean}_t \|s_t\|_2 = 0.8116$, and $\max_t \|s_t\|_2 = 0.8943 < 1.0$. ■

S1.2. Proof of Lemma 2 (Sub-Gaussian Reward Noise)

For every arm $a \in \{1, \dots, K\}$ and round t, accuracy outcome satisfies $y_{t,a} \in \{0,1\}$ and clipped normalized cost satisfies $\tilde{c}_{t,a} = \text{clip}(c_{t,a}/c_{90}, 0, 2.5) \in [0, 2.5]$. With $\gamma \ge 0$, scalar reward $r_{t,a} = y_{t,a} - \gamma \tilde{c}_{t,a}$ is almost surely bounded inside the closed interval $[-2.5\gamma, 1]$ of width $W_\gamma = 1 + 2.5\gamma$. Conditioning on history $\mathcal{F}_{t-1}$ and (s_t, a), the zero-mean noise $\eta_t = r_{t,a} - \mathbb{E}[r_{t,a} \mid s_t, \mathcal{F}_{t-1}]$ is bounded within an interval of width at most $1+2.5\gamma$. By Hoeffding's Lemma, for all $\zeta \in \mathbb{R}$:

$$\begin{aligned} &\mathbb{E}\left[\exp(\zeta \eta_t) \mid \mathcal{F}_{t-1}, s_t, a\right] \\ &\quad \le \exp\left(\frac{\zeta^2(1+2.5\gamma)^2}{8}\right) = \exp\left(\frac{\zeta^2 \nu^2}{2}\right), \end{aligned} \quad (2)$$

establishing ν-sub-Gaussianity with $\nu = \frac{1+2.5\gamma}{2}$. At our balanced setting γ=0.08, $\nu = 0.60$. ■

S1.3. Proof of Theorem 1 (Disjoint Multimodal Regret Bound & OOD Prior Decay)

Part A: High-Probability Self-Normalized Confidence Ellipsoid. Let $\mathcal{D}_{\text{off}}$ denote optional warm-start samples scaled by weight $w_0 \in [0,1]$, forming initial Gram matrix $A_a^{(0)} = \lambda I_d + w_0 \sum_{i \in \mathcal{D}_{\text{off}}, a_i = a} s_i s_i^\top$, with $\lambda_{\min}(A_a^{(0)}) \ge \lambda > 0$ and bounded eigenvalues $\lambda_{\max}(A_a^{(0)}) \le \lambda + w_0 |\mathcal{D}_{\text{off}}|$. During online steps $t = 1, \dots, T$, let T_a be the number of pulls of arm a ($\sum_{a=1}^{K} T_a = T$) and $A_{t,a} = A_a^{(0)} + \sum_{\tau < t, a_\tau = a} s_\tau s_\tau^\top$. By Theorem 2 of Abbasi-Yadkori et al. [1] applied with Lemma 1 ($\|s_t\|_2 \le 1$) and Lemma 2 (ν-sub-Gaussian noise) under stationary rewards ($\mathbb{E}[r_{t,a}|s_t] = \theta_a^\top s_t$, $\|\theta_a\|_2 \le S$), with probability at least $1-\delta$, simultaneously for all $t \ge 1$ and $a \in \{1, \dots, K\}$:

$$\|\hat{\theta}_{t,a} - \theta_a\|_{A_{t,a}} \le \beta_t(\delta/K) \le \alpha_0, \quad (2)$$

where $\beta_t(\delta/K) = \nu\sqrt{2\ln\left(\frac{K \det(A_{t,a})^{1/2}}{\delta \det(A_a^{(0)})^{1/2}}\right)} + \sqrt{\lambda + \text{bias}_{\text{off}}} S$. By Cauchy-Schwarz in the $A_{t,a}^{-1}$-induced metric:

$$\left|(\hat{\theta}_{t,a} - \theta_a)^\top s_t\right| \le \|\hat{\theta}_{t,a} - \theta_a\|_{A_{t,a}} \|s_t\|_{A_{t,a}^{-1}} \le \alpha_0 \sqrt{s_t^\top A_{t,a}^{-1} s_t}. \quad (3)$$

On the event above, the optimistic UCB choice $a_t = \arg\max_a (\hat{\theta}_{t,a}^\top s_t + \alpha_0 \|s_t\|_{A_{t,a}^{-1}})$ against hindsight optimal arm $a_t^* = \arg\max_a \theta_a^\top s_t$ satisfies instantaneous pseudo-

regret bound:

$$\begin{aligned}\mathrm{reg}_t &= \theta_{a_t^*}^\top s_t - \theta_{a_t}^\top s_t \\ &\le \hat{\theta}_{t,a_t^*}^\top s_t + \alpha_0 \|s_t\|_{A_{t,a_t^*}^{-1}} - \theta_{a_t}^\top s_t \\ &\le \hat{\theta}_{t,a_t}^\top s_t + \alpha_0 \|s_t\|_{A_{t,a_t}^{-1}} - \theta_{a_t}^\top s_t \\ &\le 2\alpha_0 \sqrt{s_t^\top A_{t,a_t}^{-1} s_t}. \end{aligned} \tag{4}$$

Since rewards lie in $[-2.5\gamma, 1]$, clipping per-step regret above by $W_\gamma = 1 + 2.5\gamma$ gives $\mathrm{reg}_t \le 2\alpha_0 \min(1, \|s_t\|_{A_{t,a_t}^{-1}})$ (for $\alpha_0 \ge W_\gamma/2$). Grouping rounds by selected arm $a \in \{1, \dots, K\}$ with counts T_a ($\sum_a T_a = T$) and applying Cauchy-Schwarz first per arm, then across the K arms via $\sum_{a=1}^K \sqrt{T_a} \le \sqrt{K \sum_{a=1}^K T_a} = \sqrt{KT}$:

$$\begin{aligned} R_T = \sum_{t=1}^T \mathrm{reg}_t &\le 2\alpha_0 \sum_{a=1}^K \sum_{t:\, a_t = a} \min\left(1, \|s_t\|_{A_{t,a}^{-1}}\right) \\ &\le 2\alpha_0 \sum_{a=1}^K \sqrt{T_a \sum_{t:\, a_t = a} \min\left(1, \|s_t\|^2_{A_{t,a}^{-1}}\right)}. \end{aligned} \tag{5}$$

By the Determinant-Trace Elliptical Potential Lemma [1] with $\det(A_{t,a}) = \det(A_a^{(0)}) \prod_{\tau: a_\tau = a} (1 + \|s_\tau\|^2_{A_{\tau,a}^{-1}})$ and Hadamard's inequality ($\mathrm{tr}(A_{T+1,a}) \le d\lambda_{\max}^{(0)} + T_a$), we obtain:

$$\begin{aligned} \sum_{t:\, a_t = a} \min(1, \|s_t\|^2_{A_{t,a}^{-1}}) &\le 2 \ln \left(\frac{\det(A_{T+1,a})}{\det(A_a^{(0)})} \right) \\ &\le 2d \ln\left(1 + \frac{T_a}{d\lambda}\right). \end{aligned}$$

Substituting back and applying $\sum_{a=1}^K \sqrt{T_a} \le \sqrt{KT}$ (plus any $\mathcal{O}(K)$ burn-in when rewards exceed initial optimism scale) establishes:

$$R_T \le 2\alpha_0 \sqrt{2dKT \ln\left(1 + \frac{T}{d\lambda}\right)} + \mathcal{O}(K) = \tilde{\mathcal{O}}(d\sqrt{KT}).$$

Part B: Why Offline Routers Fail OOD While Warm-Start Prior Decays. Consider an OOD target distribution where the true parameter vector shifts from offline source θ_a^{src} to θ_a^{ood} with suboptimality gap $\Delta_{\mathrm{shift}} > 0$ on a non-negligible fraction of OOD queries. Any frozen offline supervised classifier (Offline-Linear, ARMS, or MLP-Router) never updates weights during OOD inference, incurring strictly linear OOD regret $R_T^{\mathrm{offline}} = \Omega(\Delta_{\mathrm{shift}} T)$. In contrast, consider CascadeRoute-VL solving Ridge with offline Gram matrix scaled by $w_0 \in (0, 1)$ ($A_a^{(0)} = \lambda I_d + w_0 X_{\mathrm{src},a}^\top X_{\mathrm{src},a}$). After observing $n_{t,a}$ OOD pulls of arm a via online Sherman-Morrison updates:

$$\begin{aligned} \hat{\theta}_{t,a} - \theta_a^{\mathrm{ood}} = A_{t,a}^{-1} \Big[& w_0 X_{\mathrm{src},a}^\top X_{\mathrm{src},a} (\theta_a^{\mathrm{src}} - \theta_a^{\mathrm{ood}}) \\ & - \lambda \theta_a^{\mathrm{ood}} + \sum_{\tau \le t,\, a_\tau = a} \eta_\tau s_\tau \Big]. \end{aligned} \tag{6}$$

Since empirical OOD covariance grows linearly along active OOD directions ($\lambda_{\min}(A_{t,a}) = \Omega(\lambda + n_{t,a})$ on the spanned subspace), the stale source bias decays inversely with online sample count as $\mathcal{O}\left(\frac{w_0 \|\theta_a^{\mathrm{src}} - \theta_a^{\mathrm{ood}}\|}{\lambda + n_{t,a}}\right)$. Thus, using moderate prior weight (w_0=0.05–0.20) delivers low initial variance (∼51 Q to surpass OOD MLP) while guaranteeing rapid Bayesian prior forgetting and asymptotic convergence to θ_a^{ood}. ■

S1.4. Proof of Theorem 2 (Asymptotic Budget Feasibility)

Let $B_{\max} > 0$ denote total USD session budget and $C_t = \sum_{j=1}^t c_j$ denote cumulative spend, with remaining budget fraction $b_t = \max(0, \min(1, (B_{\max} - C_{t-1})/B_{\max}))$. First, note the Confidence Escalation Gate strictly requires $b_t > b_{\min} = 0.15$ to permit dual-arm escalation; whenever cumulative expenditure exceeds $0.85 B_{\max}$, escalation is unconditionally disabled ($c_t \le \max_a c_{t,a}$ single call per step). Second, under exploration annealing $\alpha_t = \alpha_0 b_t$, as $b_t \to 0^+$ we have $\alpha_t \to 0$ uniformly across all K arms. Consequently, exploration uncertainty bonuses vanish ($\alpha_t \sigma_a(s_t) \le \alpha_0 b_t / \sqrt{\lambda} \to 0$) and decisions converge almost surely to pure exploitation $a_t = \arg\max_a \hat{\theta}_a^\top s_t$. Since per-arm estimators regress toward cost-penalized reward $y_{t,a} - \gamma \tilde{c}_{t,a}$, setting sufficient $\gamma > 0$ steers late-session exploitation toward compact low-cost arms (∼$0.014–$0.051/1k queries), bounding late-stage spend rate so cumulative spend stabilizes prior to exceeding $B_{\max}$. ■

S2. Complete 30,540-Query Benchmark & Perceptual Data

Tables S1, S2, S3, and S4 document the complete empirical profiles reconstructed from all 238 VLMEvalKit inference files across 14 benchmarks and 17 VLMs (30,540 × 17 = 519,180 query-model pairs, > 99.9% exact cell coverage).

S3. Extended Results & Statistical Significance

Tables S5, S6, and S7 report full Pareto sweep trajectories across $\gamma \in [0, 0.9]$, Bonferroni-corrected hypothesis tests against all primary baselines, and CascadeRoute-VL's learned zero-label routing distribution across all four domains and 17 VLM arms.

Table S1. Complete Profiles of All 17 VLMs Across N=30,540 Queries. Token counts measured using cl100k_base (+256 image tokens/query); pricing follows VL-RouterBench schedules; latencies computed via class TTFT + decode throughput profiles.

Model	Params / Type	Acc (%)	Avg In Tok	Avg Out Tok	Think Tag (%)	In $/1M	Out $/1M	Cost/1k ($)	Lat (s)
Qwen2.5-VL-72B-Instruct	72B (Local)	78.21	278.3	33.3	0.0	0.80	1.20	$0.2626	2.032
InternVL2.5-78B	78B (Local)	76.63	278.3	15.8	0.0	1.00	1.50	$0.3020	1.348
Qwen2.5-VL-32B-Instruct	32B (Local)	74.50	278.3	104.4	0.0	0.40	0.60	$0.1740	3.410
Kimi-VL-A3B-Thinking-2506	3B-MoE (Local)	73.80	278.3	59.4	0.0	0.20	0.25	$0.0705	0.765
GeminiFlash2.5	API (Proprietary)	72.33	278.3	118.1	0.0	0.30	2.40	$0.3670	1.394
DeepSeek-VL2	27B (Local)	70.91	278.3	18.7	0.0	0.35	0.50	$0.1068	0.726
Qianfan-VL-8B	8B (Local)	70.61	278.3	4.3	0.0	0.18	0.25	$0.0512	0.231
DeepSeek-VL2-Tiny	1.5B (Local)	63.23	278.3	7.9	0.0	0.05	0.05	$0.0143	0.148
Gemma3-27B	27B (Local)	55.11	278.3	75.6	0.0	0.35	0.50	$0.1352	2.290
GPT-4o	API (Proprietary)	53.01	278.3	29.2	0.0	2.50	10.00	$0.9882	0.967
Phi-3.5-Vision	7B (Local)	51.62	278.3	6.0	0.0	0.10	0.10	$0.0284	0.197
SmolVLM2	2B (Local)	51.55	278.3	6.8	0.0	0.06	0.06	$0.0171	0.154
Pixtral-12B	12B (Local)	44.66	278.3	39.3	0.0	0.25	0.35	$0.0833	0.935
LLaVA-NeXT-Vicuna-7B	7B (Local)	44.54	278.3	20.2	0.0	0.20	0.20	$0.0597	0.429
MiMo-VL-7B-RL	7B (Local)	35.83	278.3	237.4	100.0	0.20	0.30	$0.1269	3.467
Janus-Pro-7B	7B (Local)	31.37	278.3	11.0	0.0	0.18	0.25	$0.0528	0.306
Janus-Pro-1B	1B (Local)	28.84	278.3	23.3	0.0	0.05	0.05	$0.0151	0.241

Table S2. Per-Dataset Unsupervised Perceptual Cues & Compact (≤8B) Rescue Breakdown Across 14 Benchmarks (N=30,540). Compact models rescue 4,557 of 6,655 Qwen2.5-VL-72B failures (68.5% overall) using zero domain labels.

Dataset	Domain	N	Ent (b)	Sobel	WhiteBg	Words	MathCue	Oracle	72B	Best ≤8B (Acc)	72B Fails Rescued
AI2D_TEST	STEM	3,088	3.34	0.162	0.710	9.6	0.005	99.68	88.96	Qianfan-8B (85.04)	90.9%
MathVerse_MINI	STEM	3,940	2.00	0.117	0.887	39.8	0.359	90.10	51.50	MiMo-7B-RL (66.50)	67.5%
MathVision_MINI	STEM	304	2.39	0.133	0.835	61.6	0.292	92.11	38.82	Kimi-Think (47.04)	69.4%
MathVista_MINI	STEM	1,000	3.72	0.136	0.600	41.4	0.268	96.10	73.90	Kimi-Think (79.60)	75.9%
MMBench_DEV_EN	General	1,292	6.28	0.157	0.162	8.9	0.002	97.91	88.24	Qianfan-8B (84.67)	72.4%
MMMU_DEV_VAL	General	1,050	3.91	0.175	0.578	33.0	0.269	97.33	67.90	Kimi-Think (60.48)	83.1%
MMStar	General	1,500	5.11	0.152	0.372	16.8	0.029	96.73	71.40	MiMo-7B-RL (70.87)	81.8%
RealWorldQA	General	765	6.94	0.142	0.005	8.4	0.016	99.35	75.42	MiMo-7B-RL (72.55)	87.2%
HallusionBench	Halluc.	951	3.93	0.169	0.522	16.2	0.123	98.74	72.87	Kimi-Think (72.98)	92.6%
ChartQA_TEST	OCR	2,500	3.15	0.095	0.761	11.4	0.236	97.12	87.20	Qianfan-8B (86.48)	70.6%
DocVQA_VAL	OCR	5,349	3.69	0.177	0.485	8.5	0.065	98.47	91.21	Qianfan-8B (87.42)	71.7%
InfoVQA_VAL	OCR	2,801	5.54	0.281	0.309	11.1	0.110	96.14	82.86	Qianfan-8B (67.37)	53.1%
OCRBench	OCR	1,000	5.32	0.218	0.296	10.3	0.044	98.50	88.70	Qianfan-8B (85.70)	72.6%
TextVQA_VAL	OCR	5,000	7.10	0.163	0.054	7.1	0.009	89.44	75.82	Qianfan-8B (73.78)	48.5%

S4. Qualitative Reasoning Tax Gallery & Reproducibility

Figure S1 displays four authentic benchmark queries extracted verbatim from VL-RouterBench transcripts illustrating both failure modes CascadeRoute-VL resolves:

- OCRBench #314 (Yellow Cab): Ground truth "Y809". Qianfan-8B reads "The cab number is Y809." (7 tok, right); Qwen2.5-72B (17 tok), Kimi-Think (39 tok), and MiMo-VL-RL (136 tok) confuse the fleet phone "426-6262" or fabricate "7007".
- OCRBench #979 (LaTeX Inequality): Ground truth "$m \neq \sqrt{2}$". Qianfan-8B outputs "m \neq \sqrt { 2 }" (8 tok, right); Qwen2.5-72B, Kimi-Think, and DS-Tiny misread ≠ as +, while MiMo-VL-RL emits 189 speculative <think> tokens ("$M + \sqrt{x}$").
- MathVista #897 & #874 (Trigonometry & Step-Function): Compact perception arms guess wrong in 1–5 tokens and Qwen2.5-72B misidentifies cos vs. sin (197 tok, wrong) and step-interval endpoints (157 tok, wrong), while Kimi-VL-A3B-Thinking derives exact answers ("40" in 178 tok; "2" in 269 tok).

S4.1. Reproducibility & Implementation Details

All experiments execute in Python 3.13.5 + PyTorch CUDA (RTX 4050 GPU) with offline local HuggingFace/TorchHub weights (all-MiniLM-L6-v2, DINOv2-ViT-S/14, ResNet-50). Feature normalization weights (w_q, w_v, w_p)=$(0.50, 0.55, 0.235)$ are fixed analytically by Lemma 1. Bandit hyperparameters (α_0=0.10, λ=4.0, w_0=0.20; confidence gate τ_u=0.12, τ_m=0.025, p_{cap}=0.78, $b_{\min}$=0.15, cost penalty differential κ=0.22) were calibrated exclusively on the Dev split (N=3,054), and all Test (N=6,108) and OOD (N=17,601) streaming metrics are reported as mean ± std over 5 random permutations (seeds 42–46). Full unit tests (pytest tests/ -v: 45/45 passing) verify Sherman-Morrison exactness, zero-leakage state norms, and Confidence Gate escalation constraints.

Table S3. Full 14 × 17 Benchmark Accuracy Matrix (%). Abbreviations: GPT4o, Gem2.5 (GeminiFlash2.5), Gem27 (Gemma3-27B), Int78 (InternVL2.5-78B), J1B/J7B (Janus-Pro-1B/7B), Kimi (Kimi-VL-A3B-Thinking), MiMo (MiMo-VL-7B-RL), Phi3.5, Pix12 (Pixtral-12B), Qian8 (Qianfan-VL-8B), Q32/Q72 (Qwen2.5-VL-32B/72B), Smol2 (SmolVLM2), DS27/DST (DeepSeek-VL2/Tiny), LLaVA7.

Dataset	GPT4o	Gem2.5	Gem27	Int78	J1B	J7B	Kimi	MiMo	Phi3.5	Pix12	Qian8	Q32	Q72	Smol2	DS27	DST	LLaVA7	Oracle
AI2D	83.2	88.6	81.4	88.9	66.9	68.0	82.6	83.3	77.6	44.7	85.0	85.6	89.0	69.6	82.8	74.6	66.8	99.7
ChartQA	36.7	38.7	30.0	86.0	23.2	25.7	73.0	0.0	71.2	63.3	86.5	70.2	87.2	68.6	85.8	81.1	54.0	97.1
DocVQA	38.5	85.9	56.1	89.1	0.6	1.1	87.1	0.0	54.1	59.0	87.4	87.5	91.2	63.9	85.1	80.1	50.2	98.5
Hallusion	69.1	76.6	64.0	72.3	51.1	60.3	73.0	72.3	59.3	46.4	69.3	72.1	72.9	60.2	64.4	61.6	47.2	98.7
InfoVQA	28.6	77.4	28.4	77.3	0.0	0.0	66.3	0.0	25.1	18.5	67.4	76.4	82.9	0.0	67.5	55.6	20.0	96.1
MMBench	82.2	89.6	80.4	89.2	58.7	64.5	84.0	84.1	71.8	60.8	84.7	86.5	88.2	67.4	82.5	70.9	65.6	97.9
MMMU	68.1	77.9	62.8	66.5	37.1	42.0	60.5	59.5	44.3	32.5	48.4	69.9	67.9	41.6	54.2	42.0	38.5	97.3
MMStar	61.1	77.0	60.6	69.5	43.1	46.1	69.9	70.9	47.2	41.4	63.3	69.7	71.4	45.9	62.2	49.5	37.8	96.7
MathVerse	39.1	50.8	41.4	48.5	17.0	20.9	61.0	66.5	22.9	15.8	31.1	54.5	51.5	22.5	37.0	23.9	13.4	90.1
MathVision	32.2	59.2	33.2	30.3	16.8	17.1	47.0	35.9	16.8	15.8	18.1	36.5	38.8	17.4	22.0	16.1	14.5	92.1
MathVista	57.7	80.2	60.6	70.6	35.7	43.0	79.6	77.8	43.7	41.8	64.0	73.3	73.9	51.5	63.7	53.8	31.5	96.1
OCRBench	72.0	85.6	75.1	85.0	49.0	59.3	82.8	84.6	60.3	53.0	85.7	85.4	88.7	72.9	81.3	80.3	50.4	98.5
RealWorldQA	68.5	76.7	62.5	78.4	54.8	42.1	70.9	72.6	53.9	63.8	70.5	69.8	75.4	57.5	69.9	64.7	58.6	99.4
TextVQA	60.7	66.7	59.9	76.8	37.3	40.4	69.5	0.0	58.5	54.1	73.8	71.4	75.8	65.4	76.5	72.5	56.8	89.4

Table S4. Full 14 × 17 Mean Generated Output Token Matrix (cl100k_base). Notice how MiMo-VL-7B-RL emits up to 1,401.7 output tokens on MathVision yet collapses to 4.8–55.7 broken tokens on OCR datasets, whereas Qianfan-VL-8B answers OCR in 1.2–4.6 tokens.

Dataset	GPT4o	Gem2.5	Gem27	Int78	J1B	J7B	Kimi	MiMo	Phi3.5	Pix12	Qian8	Q32	Q72	Smol2	DS27	DST	LLaVA7
AI2D	13.7	233.2	97.9	1.0	1.0	1.4	113.5	273.7	1.1	33.8	1.0	131.7	3.7	1.0	19.1	1.0	1.0
ChartQA	1.5	4.8	2.2	1.2	1.2	1.2	14.7	44.8	1.2	1.2	1.2	1.5	1.2	1.2	1.2	1.2	1.2
DocVQA	2.1	4.2	2.6	2.8	3.0	2.5	5.0	55.7	2.9	2.2	2.7	2.9	2.9	2.8	2.8	2.7	2.8
Hallusion	50.8	75.2	77.9	1.0	1.5	1.0	70.6	335.4	18.9	42.7	24.1	164.2	58.6	1.0	23.6	9.6	64.2
InfoVQA	1.6	2.4	2.0	1.9	2.6	1.7	22.1	5.5	1.6	1.2	1.9	2.9	2.0	1.8	1.9	2.6	2.0
MMBench	12.0	169.2	104.1	1.0	1.0	1.8	67.4	269.1	1.1	32.1	1.0	63.6	7.1	1.0	2.5	1.2	1.0
MMMU	173.9	520.1	321.9	15.6	3.9	28.5	2.3	717.3	12.0	113.8	5.1	419.0	32.0	6.8	20.2	4.8	7.6
MMStar	58.3	261.3	152.3	1.0	1.1	3.4	1.2	377.5	1.3	60.2	1.0	152.2	4.9	1.0	9.7	1.1	1.3
MathVerse	56.2	249.9	198.2	40.0	160.1	59.9	209.7	702.1	20.6	147.6	2.1	347.7	144.6	37.8	102.6	43.6	74.3
MathVision	355.0	846.6	554.9	367.5	23.5	5.0	346.3	1401.7	33.4	216.7	135.9	521.4	375.0	4.2	5.0	5.0	195.2
MathVista	135.8	264.8	176.0	153.6	17.4	7.1	142.5	491.1	4.1	107.0	9.1	245.2	156.8	4.8	5.0	5.2	113.6
OCRBench	17.7	19.5	32.2	1.9	9.3	14.7	46.0	125.8	18.9	17.3	4.6	41.7	19.9	4.2	7.0	6.2	41.7
RealWorldQA	7.5	106.6	52.2	1.0	1.0	1.1	19.3	209.3	1.1	9.8	1.1	27.1	6.2	4.2	1.0	1.0	1.0
TextVQA	1.7	2.2	1.9	1.8	1.7	2.2	8.8	4.8	1.7	1.4	1.7	2.2	1.9	1.7	1.8	1.8	1.7

Table S5. Full Pareto Sweep Data Corresponding to Main Paper Figure 3 (Test N=6,108). CascadeRoute-VL maintains smooth monotonic accuracy-cost trade-offs across the entire cost spectrum.

Method (γ)	Acc (%)	Cost/1k	Tok	Lat (s)	Method (γ/τ)	Acc (%)	Cost/1k	Tok	Lat (s)
CascadeRoute-VL (0.00)	80.42	$0.2840	395.7	1.959	KNN-Router (0.00)	80.29	$0.2812	369.6	1.667
CascadeRoute-VL (0.03)	79.99	$0.2316	388.4	1.840	KNN-Router (0.08)	78.31	$0.1306	361.8	1.444
CascadeRoute-VL (0.08)	78.49	$0.1585	368.6	1.516	KNN-Router (0.30)	75.95	$0.0753	344.4	1.091
CascadeRoute-VL (0.15)	77.44	$0.1115	359.2	1.180	KNN-Router (0.90)	73.05	$0.0431	320.6	0.653
CascadeRoute-VL (0.30)	75.93	$0.0879	360.4	0.831	ARMS-Router (0.00)	80.14	$0.3063	371.0	1.411
CascadeRoute-VL (0.55)	75.33	$0.0844	377.4	0.699	ARMS-Router (0.08)	78.80	$0.2190	369.0	1.306
CascadeRoute-VL (0.90)	74.36	$0.0818	410.1	0.672	ARMS-Router (0.90)	72.22	$0.0448	321.3	0.589
MLP-Router (0.00)	81.01	$0.2834	406.3	2.068	Offline-Linear (0.00)	79.68	$0.2718	386.0	1.817
MLP-Router (0.03)	80.47	$0.2399	406.0	2.006	Offline-Linear (0.08)	78.21	$0.1777	384.3	1.721
MLP-Router (0.08)	79.85	$0.1854	400.4	1.844	Offline-Linear (0.90)	71.37	$0.0470	328.3	0.715
MLP-Router (0.15)	78.65	$0.1412	390.0	1.656	FrugalGPT (τ=0.55)	69.38	$0.0641	472.9	0.777
MLP-Router (0.30)	76.31	$0.0883	365.9	1.289	FrugalGPT (τ=0.65)	71.12	$0.1058	579.2	1.184
MLP-Router (0.55)	74.79	$0.0588	345.4	0.984	FrugalGPT (τ=0.72)	73.23	$0.1704	725.8	1.719
MLP-Router (0.90)	72.69	$0.0470	330.9	0.761	FrugalGPT (τ=0.88)	76.74	$0.3515	1121.1	2.906

Table S6. Statistical Significance Summary (Test N=6,108, Paired Sample Tests on Canonical Stream Seed 42 vs. CascadeRoute-VL γ=0.08, Single-Seed Acc 78.49%).

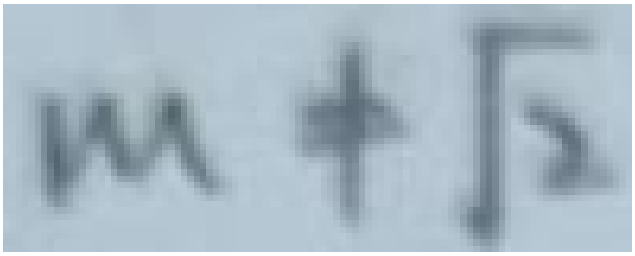
Comparison	Acc Δ (%)	Acc t	Acc p-val	Wilcoxon p	Cohen's d	USD Saved (/1k)	Cost p-val
vs. Qwen2.5-VL-72B	+1.20	2.433	1.50×10^{-2}	1.50×10^{-2}	+0.031	+$0.1021	9.49×10^{-275}
vs. MLP-Router (i.i.d. static)	-1.36	-3.514	4.45×10^{-4}	4.47×10^{-4}	-0.045	+$0.0269	6.30×10^{-12}
vs. ARMS-Router	-0.31	-0.723	4.70×10^{-1}	4.70×10^{-1}	-0.009	+$0.0605	3.67×10^{-41}
vs. Offline-Linear	+0.28	0.659	5.10×10^{-1}	5.10×10^{-1}	+0.008	+$0.0192	2.30×10^{-8}
vs. FrugalGPT-Cascade	+5.26	10.605	4.78×10^{-26}	7.85×10^{-26}	+0.136	+$0.0119	1.67×10^{-4}
vs. LinUCB (Cold Online)	+5.98	10.041	1.53×10^{-23}	2.28×10^{-23}	+0.129	+$0.1937	5.06×10^{-173}
vs. MHR-Router	+6.57	12.010	7.39×10^{-33}	1.67×10^{-32}	+0.154	-$0.0189	1.44×10^{-7}

Table S7. Domain Breakdown & 17-Arm Selection Distribution (Test N=6,108, Zero Task Labels). Unlisted compact/legacy arms (GPT-4o, Janus-1B/7B, LLaVA-7B, Pixtral-12B) receive 0.0% selection.

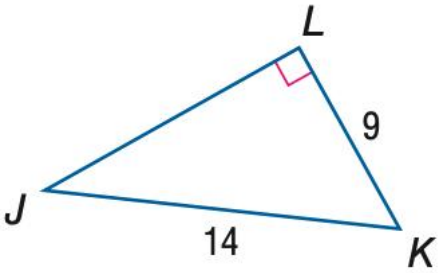
Domain (Test N)	72B Acc	Kimi Acc	Qianfan	Ours Acc	Ours OutTok	Active Arm	Overall (%)	STEM (%)	OCR (%)
STEM (1,631)	67.08	70.39	54.75	74.62	259.6	Qwen2.5-VL-72B	19.52	6.99	26.09
General_VQA (892)	73.21	69.84	66.48	73.77	102.7	Kimi-3B-Think	19.47	36.79	8.47
Charts_OCR (3,400)	83.56	75.00	78.29	81.91	4.9	Qianfan-VL-8B	17.31	4.66	26.03
Hallucination (185)	71.89	69.73	64.86	72.43	90.1	GeminiFlash2.5	11.10	10.97	12.06
						DeepSeek-VL2 (27B)	9.87	2.21	14.62
						MiMo-VL-7B-RL	8.97	26.36	0.09
						Qwen2.5-VL-32B	5.62	6.01	5.35
						InternVL2.5-78B	4.86	4.90	2.59
						DeepSeek-Tiny / Smol2	2.84	0.06	4.59



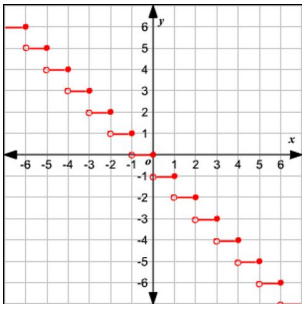
(a) OCRBench #314 (GT: Y809) — Qianfan-8B: Correct (7 tok); 72B/Think: Wrong.



(b) OCRBench #979 (GT: $m \neq \sqrt{2}$) — Qianfan-8B: Correct (8 tok); 72B/Think: Wrong.



(c) MathVista #897 (GT: 40) — Kimi-Think: Correct (178 tok); Direct/72B: Wrong.



(d) MathVista #874 (GT: 2) — Kimi-Think: Correct (269 tok); Direct/72B: Wrong.

Figure S1. Authentic VL-RouterBench Qualitative Gallery. CascadeRoute-VL learns to pair compact perception models with high-frequency scene text/formulas while routing multi-step STEM diagrams to reasoning specialists.

References

[1] Yasin Abbasi-Yadkori, Dávid Pál, and Csaba Szepesvári. Improved algorithms for linear stochastic bandits. In NeurIPS, 2011. 1, 2